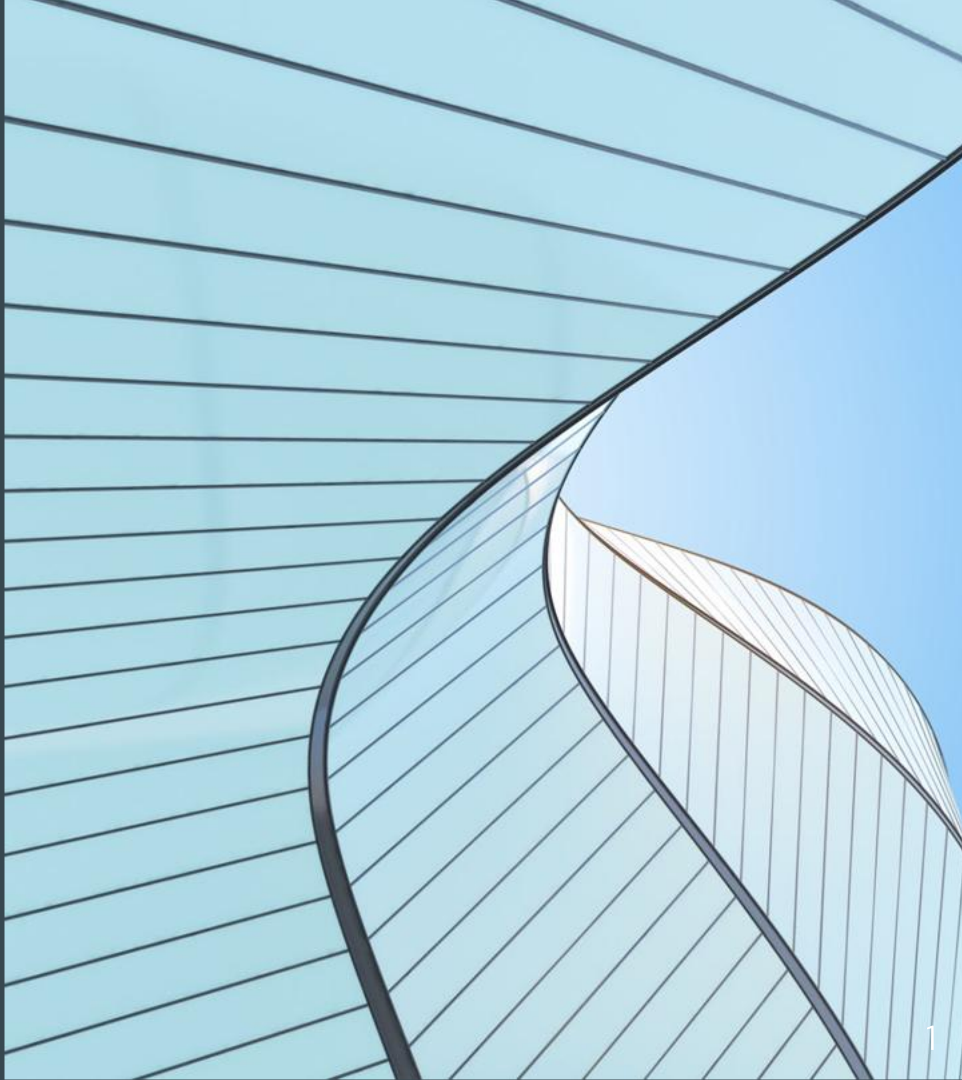
Tutoring Center Infrastructure Proposal



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Project Purpose & Scope

Purpose:

Establish secure, efficient, and scalable infrastructure across three tutoring centers

Scope:

- Network Theory and Background
- Physical and Network topologies
- Equipment Selection
- Security Methods: Firewalls, VPNs, VLANs, etc.
- Budget and cost estimates
- Ongoing IT Support Strategies

Theory and Background (Network Fundamentals)

- Local Area Network (LAN) - Local network (office/home)
- Metropolitan Area Network (MAN) - City-wide network (campus or city)
- Wide Area Network (WAN) - Connect LANs over large distances
- Internet - Global network of interconnected systems using TCP/IP to facilitate a connection

Theory and Background (Network Technologies)

- LAN Tech: Ethernet, Wi-Fi, Fiber Optics
- WAN Tech: MPLS, SD-WAN, Leased Lines
- Wireless: Wi-Fi, LTE, Bluetooth
- Security Mechanisms: Firewalls, IDS/IPS, VLAN
- VPN: Encrypted tunnels used across public networks

OSI Model Overview - A 7 Layer Framework

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

OSI Layers 1-3:

- Layer 1 - Physical: Bits, cables, radio, NIC (network interface cards)
- Layer 2 - Data Link: MAC addressing, switches, frames
- Layer 3 - Network: IP addressing, routers, routing protocols

OSI Model Overview - A 7 Layer Framework (continued)

OSI Layers 4-7

- Layer 4 - Transport: TCP/UDP, segmentation, error recovery
- Layer 5 - Session: Establishes and manages sessions
- Layer 6 - Presentation: Format translation, encryption, and compression
- Layer 7 - Application: HTTP, FTP, SMTP, DNS

TCP/IP Model Overview - A 4 Layer Framework

- Link Layer: Local delivery, MAC addressing (Ethernet, Wi-Fi)
- Network Layer: IP addressing and routing (IPv4/IPv6, ICMP)
- Transport Layer: TCP (reliable, connection-oriented) / UDP (fast, connectionless, lacks reliability)
- Application Layer: HTTP, FTP, DNS, SMTP

Common Security Threats

- DDos Attacks: Flood servers with traffic → slowdowns or outages
  - Defense: Firewalls, Intrusion Prevention System (IPS), traffic anomaly detection
- Phishing Attacks: Deceptive emails to trick users into sharing confidential information (usernames/passwords)
  - Defense: Email filtering, user training, multi-factor authentication (MFA)
- Man-in-the-Middle (MitM): Attackers intercept communications
  - Defense: SSL/TLS encryption, VPNs
- Malware & Ransomware: Steal or encrypt data, may also demand ransom
  - Defense: Antivirus, backups, network segmentation

Security Mechanisms and Defensive Strategies

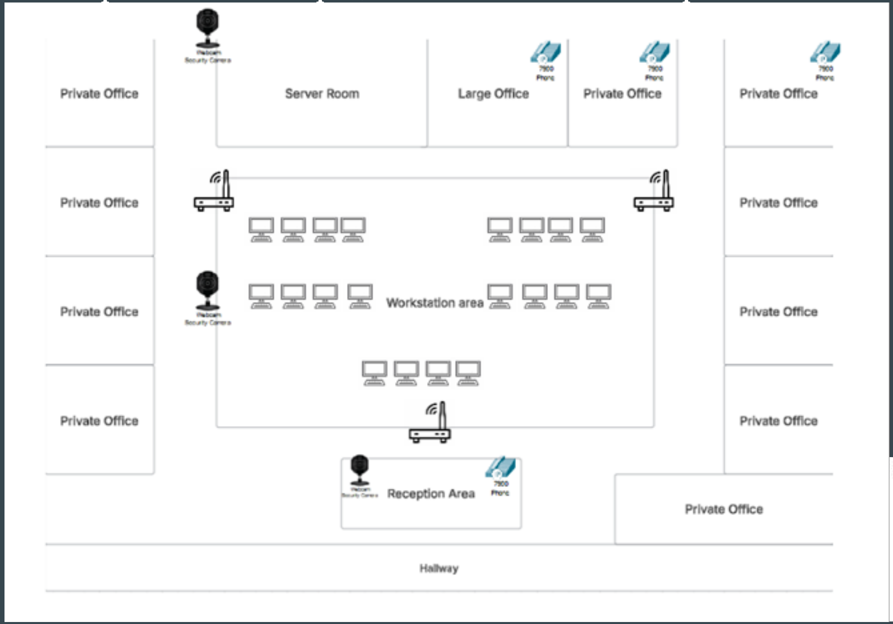
- Firewalls - Block unauthorized traffic using preset security protocols
  - Network Firewalls: Cisco ASA, Palo Alto
- Encryption - Converts data into an unreadable format to protect confidentiality
  - SSL/TLS : Encrypts web traffic (HTTPS)
  - AES: Secures stored/transmitted data
- IDS/IPS - Detects or blocks suspicious activity
  - Intrusion Detection System (IDS): Detects threats (Snort, Suricata)
  - Intrusion Prevention System (IPS): Blocks threats in real time
- VLAN Segmentation - Separates network traffic to limit access
  - Isolates traffic → limits lateral movement for intruders

Content Filtering and Legal Compliance

- Content Filtering Tools
  - DNS Filtering: Block malicious domains
  - Web Proxies: Prevents access to restricted websites
- Illinois & Federal Regulations
  - PIPA (Illinois Personal Information Protection Act): Requires encryption of personal data
  - BIPA (Biometric Information Privacy Act): Protects the biometric data from unauthorized use
  - GLBA (Gramm-Leach-Bliley Act) - Regulates financial data security
  - CISA (Cybersecurity Information Sharing Act) - Encourages organizations to report cyber threats

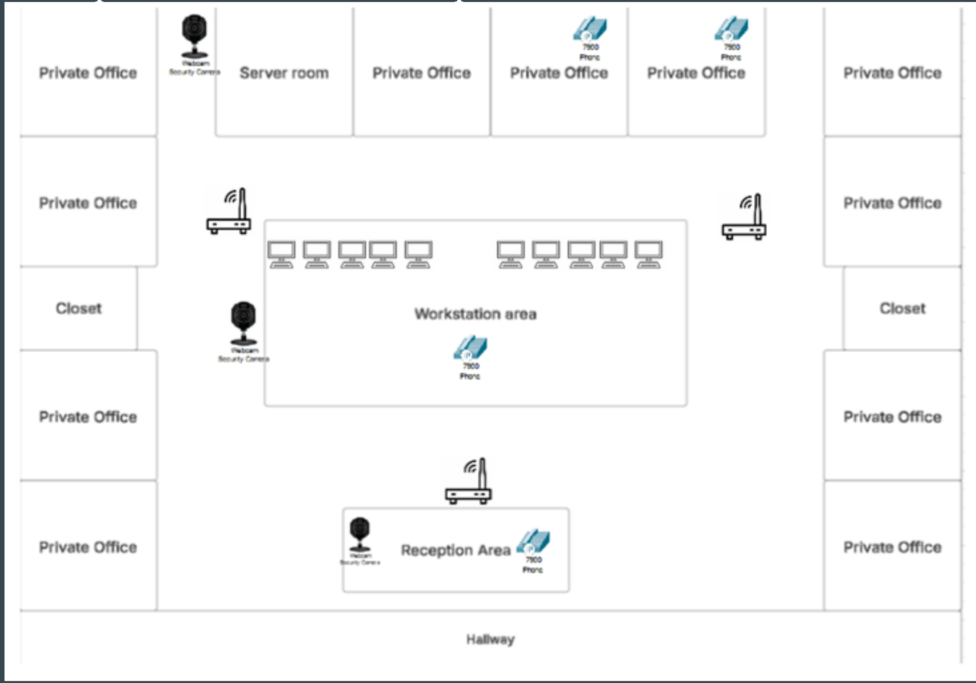
Romeoville, IL Tutoring Center (Physical Layout Overview)

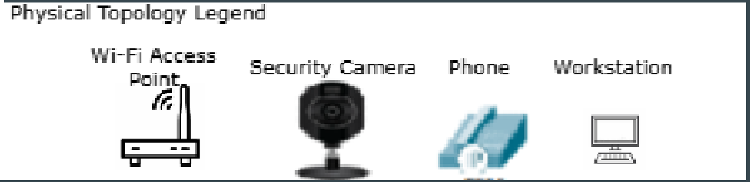
Romeoville, IL (Main Center - 3,600 sq ft): Reception area (400 sq ft), 1 large office (200 sq ft), 10 private offices (150 sq ft each), open workspace area (1200 sq ft), and a server room (400 sq ft)



Chicago, IL Tutoring Center (Physical Layout Overview)

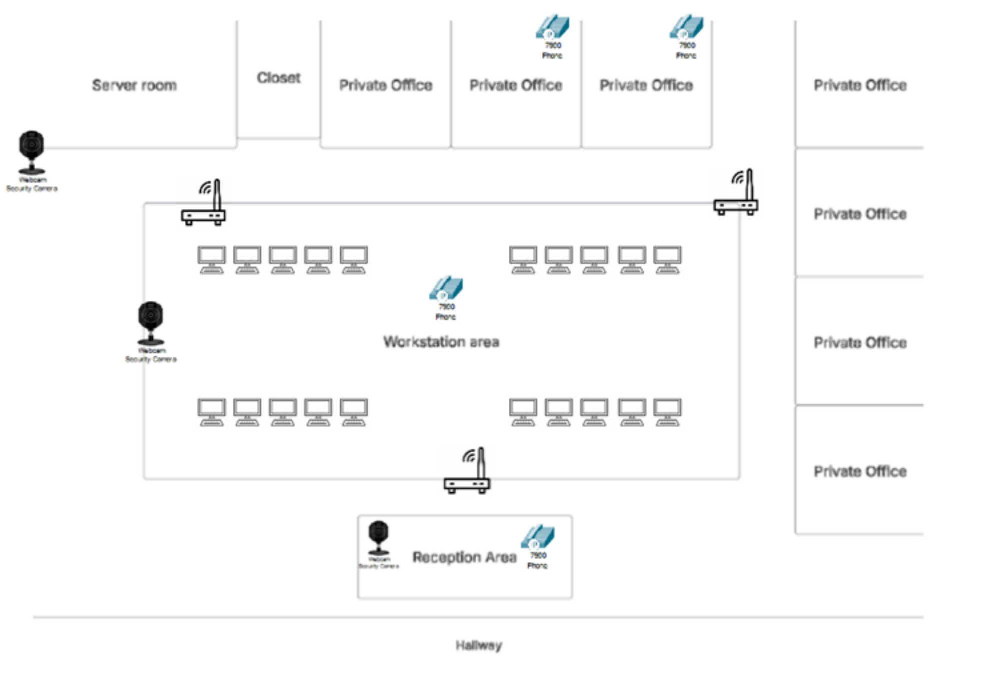
Chicago, IL (3,000 sq ft): Reception area (400 sq ft), 11 private offices (150 sq ft each), open workspace area (600 sq ft), server room (150 sq ft), and two closets (100 sq ft each)

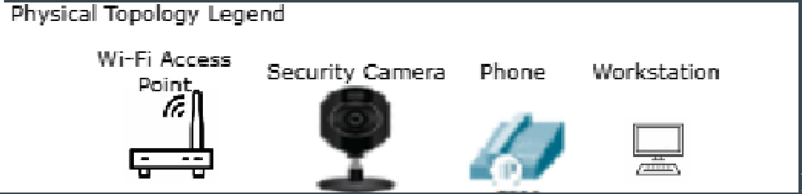




Aurora, IL Tutoring Center (Physical Layout Overview)

Aurora, IL (3,000 sq ft): Reception area (400 sq ft), 7 private offices (150 sq ft each), open workspace area (1200 sq ft), s erver room (200 sq ft), and a closet (150 sq ft)





Installation Considerations (Wiring runs) - Romeoville, IL

- Drop ceilings and wall pass-throughs - All data and power cables will run above the drop ceilings and can be guided where needed
- Server room wiring trunk - Fiber backbone (plenum-rated if in ceiling spaces) connecting all areas such as the reception area, privates offices, open-workspaces, and required devices
- Scheduled off-peak installations - All new cable installations should occur during non-operation hours or maintenance windows to reduce interruptions
- Wiring Required:
  - 3000 ft of Cat6a
  - 100-300 ft of fiber
  - Extra 100ft of Cat6a as patch cables

Installation Considerations (Wiring runs) - Chicago, IL

- Combination of ceiling and closet runs - Using plenum-rated fiber connections, network is connected via cat6a cables extending toward offices, open-workspace, and required devices
- Local code compliance - Cable pathways must meet local building codes and TIA/EIA standards for structured cabling
- Minimized downtime - Where possible, existing conduits and cable trays may be reused to reduce construction time and downtime
- Wiring Required:
  - 2500 ft of Cat6a
  - 100-200 ft of fiber
  - Extra 100ft of Cat6a as patch cables

Installation Considerations (Wiring runs) - Aurora, IL

- Raised floor infrastructure - Using riser-rated fiber cables ran below the floor to link all required areas to the server room
- Secure cable management - Cables are ran in conduit or cable trays beneath the floor, and protecting them from potential damage
- Maintenance windows - Newer installation or rerouting are to be scheduled during downtime to minimize interruptions
- Wiring Required:
  - 2500 ft of Cat6a
  - 100-200 ft of fiber
  - Extra 100ft of Cat6a as patch cables

Equipment Relocation/Disposal

- Legacy devices or wiring should be safely relocated, and outdated equipment will be recycled/disposed of following the state and federal regulations
  - Resource Conservation and Recovery Act (RCRA): Federal law that governs the disposal of hazardous waste (lead, mercury or cadmium) found in electronics
  - Occupational Safety and Health Administration (OSHA): Safety standards to ensure safe work practices
  - Privacy Regulations: Following the FTC guidelines for the destruction and disposal of old devices that may contain sensitive information

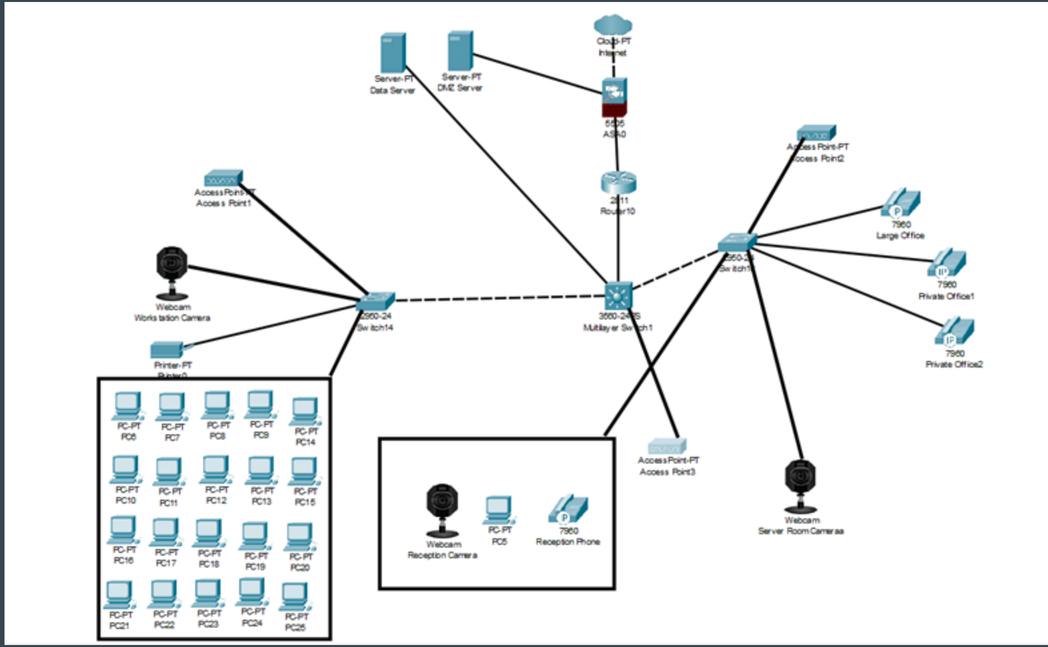
Physical Security

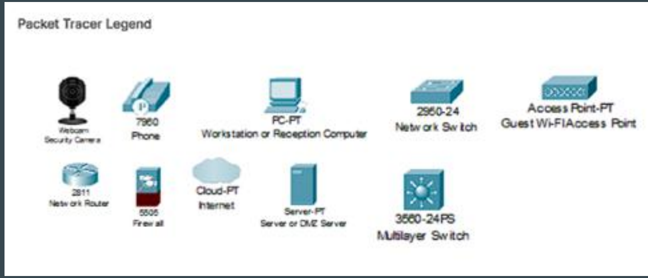
- Critical network devices (routers, switches, or servers) will be properly secured in a locked server room or secure racks with proper environmental controls
- Devices within the server room are installed onto lockable racks with built-in mechanical door locks as an extra defensive measure
- Wireless Access Points (WAP) will be mounted in secure locations around the tutoring centers to ensure full coverage

LAN Architecture

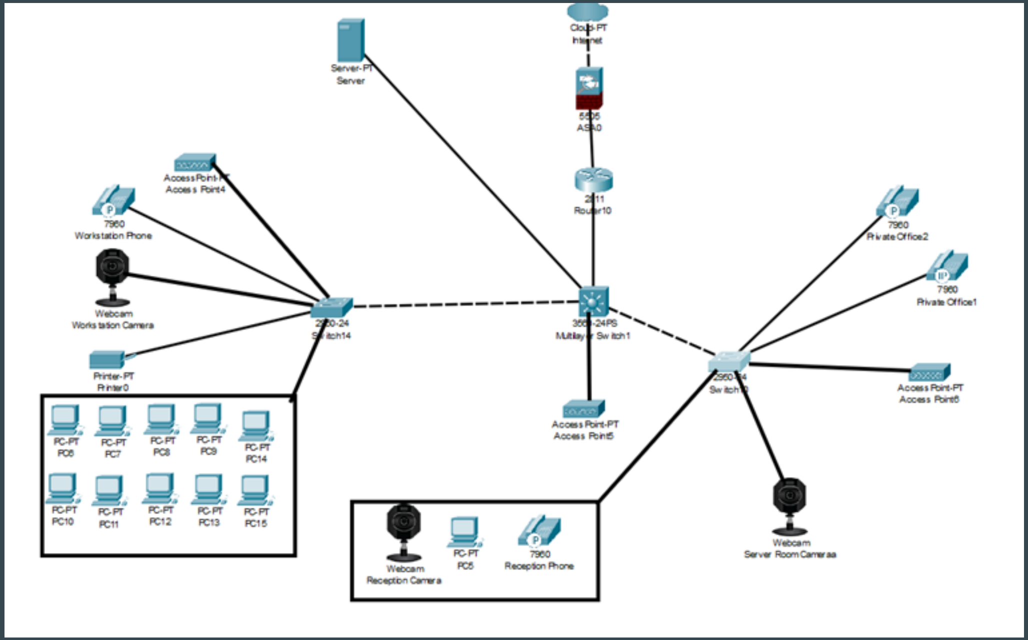
- Internet Connection - High-speed internet connection to each tutoring center
- Router/Firewall - Secures and connects via fiber uplinks, and establishes a IPsec VPN tunnels to other tutoring centers
- Managed Switch - Distributes traffic along the network and supports VLAN segmentation
- Fiber Uplinks:
  - Plenum-rated cables run along the ceiling (Romeoville/Chicago)
  - Riser-rated cables run below the floor (Aurora)

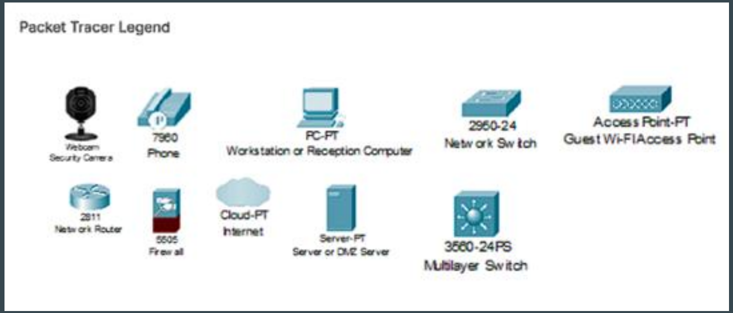
Romeoville, IL (Network Topology Overview)



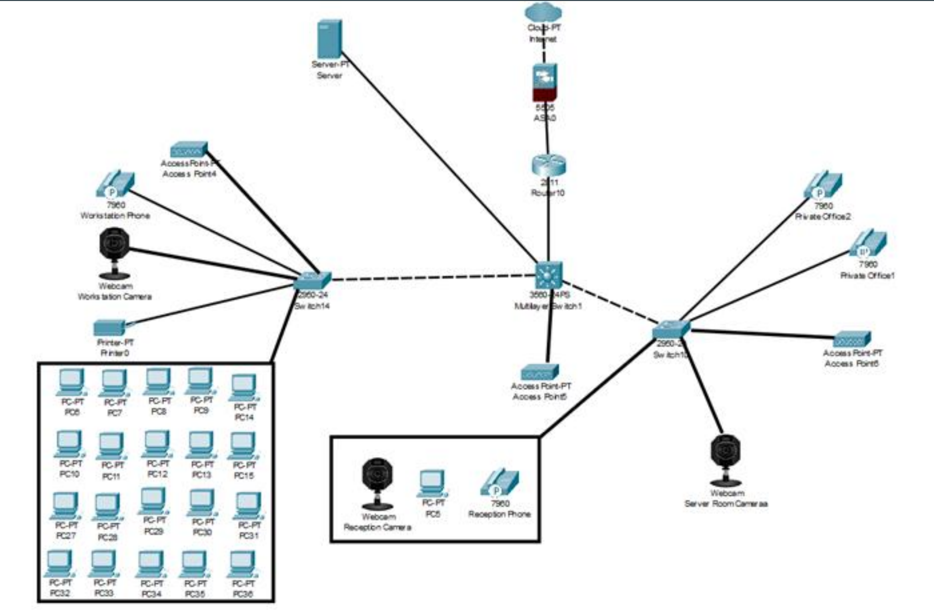


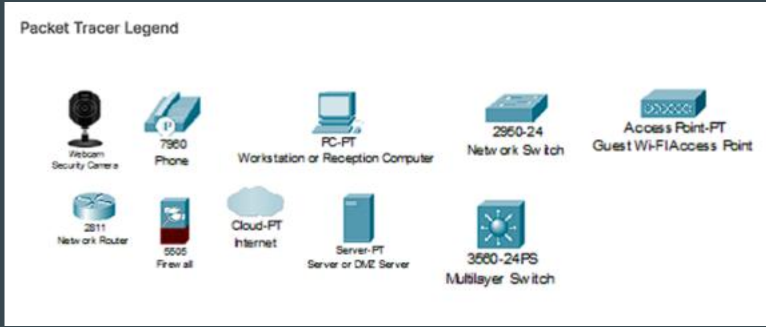
Chicago, IL (Network Topology Overview)





Aurora, IL (Network Topology Overview)





Network Equipment Installation

Location	Key Equipment
Romeoville	DMZ Server, Storage Server, Firewall, Router, Switches, Wireless APs, UPS
Aurora	Firewall, Router, Switch, Wireless APs, UPS
Chicago	Firewall, Router, Switch, Wireless APs, UPS

Security and Compliance Implementation

Security Devices for Network Protection Device Comparison

- Firewalls
- Security Incident and Event Monitoring (SIEM)
- Intrusion Detection & Prevention Systems IDP/IPS
- Data Loss Prevention (DLP)
- Web Filtering
- Web Authentication

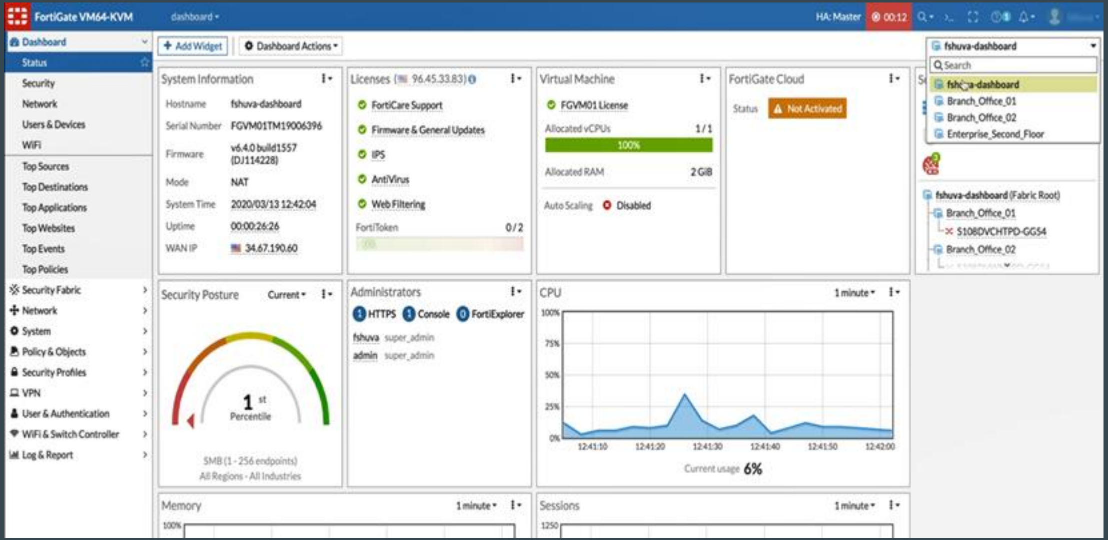
Firewalls

Prevent unauthorized access and filter traffic at the network perimeter.

3 Options:

- Fortinet FortiGate 100F
- Cisco Meraki MX85
- Palo Alto Networks PA-220

Fortinet FortiGate 100F



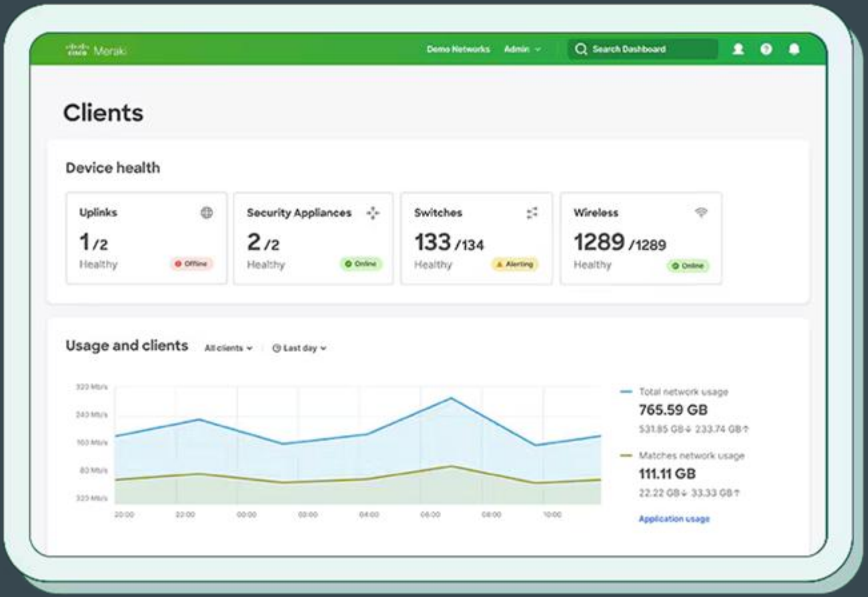
Pros:

- Cost-Effective for medium enterprises and branch offices
- Built-in UTM (Unified Threat Management): IPS, antivirus, application control
- High VPN performance which is crucial for branch-to-branch tunneling

Cons:

- UI can have a slight learning curve
- Rackmount size might be to big for smaller sites

Cisco Meraki MX85



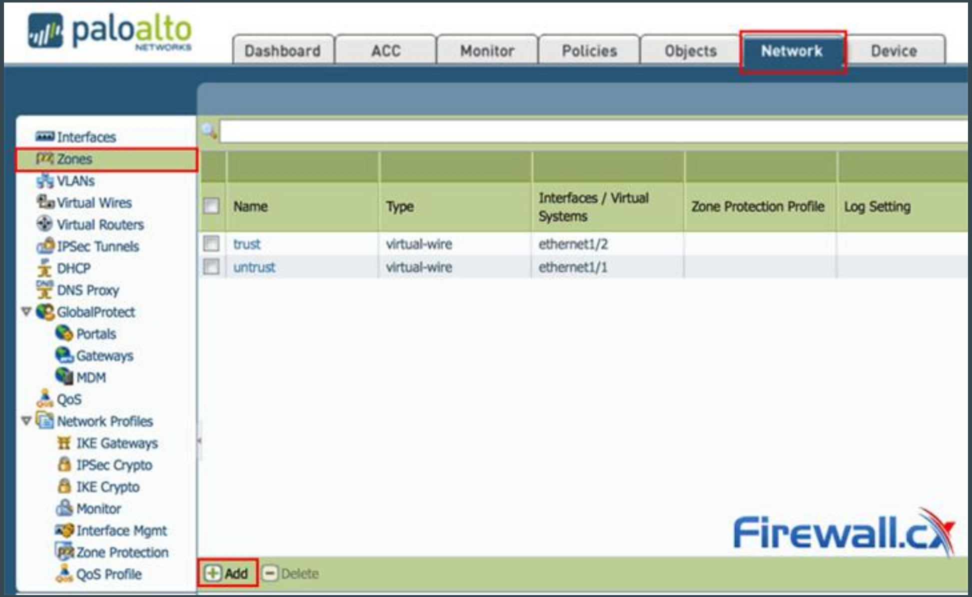
Pros:

- Simplified Cloud Management that manages everything via Meraki Dashboard
- Auto VPN setup for site-to-site VPNs
- Automatic firmware upgrades and threat signature updates

Cons:

- High recurring licensing cost
- Performance bottlenecks under heavy loads
- Cloud dependency

Palo Alto Networks PA-220



Pros:

- Best-in-class threat intelligence
- Built-in support for Zero Trust security models
- High visibility into application usage and risks

Cons:

- More complex initial configuration
- Lower throughput meaning not suitable for sites needing high VPN bandwidth

Firewall Final Recommendation

- Primary Recommendation: Fortinet FortiGate 100F is the best combination of cost, performance, VPN, and security.

- Alternate Recommendation: Cisco Meraki MX85 simplified, cloud based, but can be pricey and experience bottlenecks.

- Palo Alto PA-220 would only be recommended if you had very strict compliance or application filtering requirements.

Security Information and Event Management (SIEM)

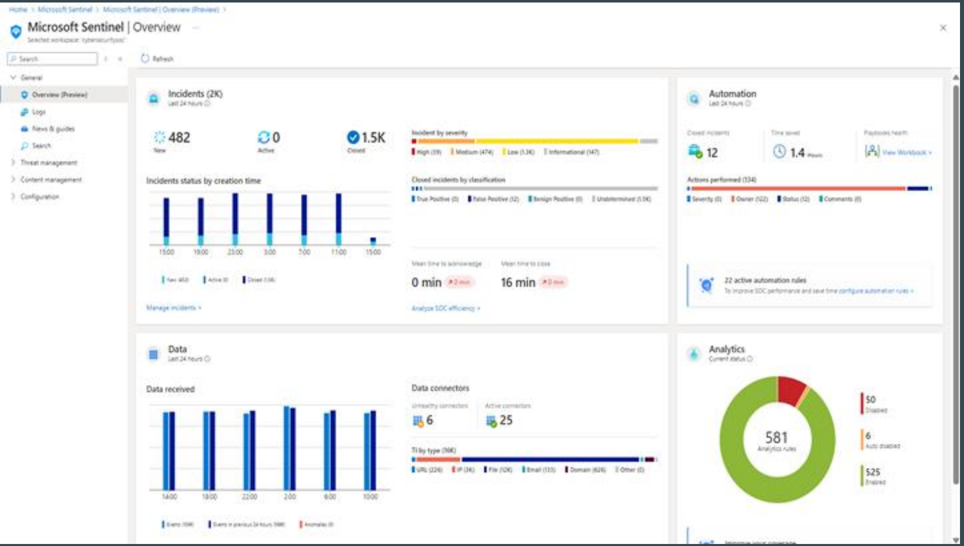
Aggregate logs and security events from multiple devices, perform real-time analysis, and alerts administrators of threats.

- Analyze traffic patterns
- Correlate events across sites
- Detect threats in real time
- Notify administrators of suspicious behavior.

3 Options

- Microsoft Sentinel
- Splunk Enterprise Security
- IBM QRadar

Microsoft Sentinel



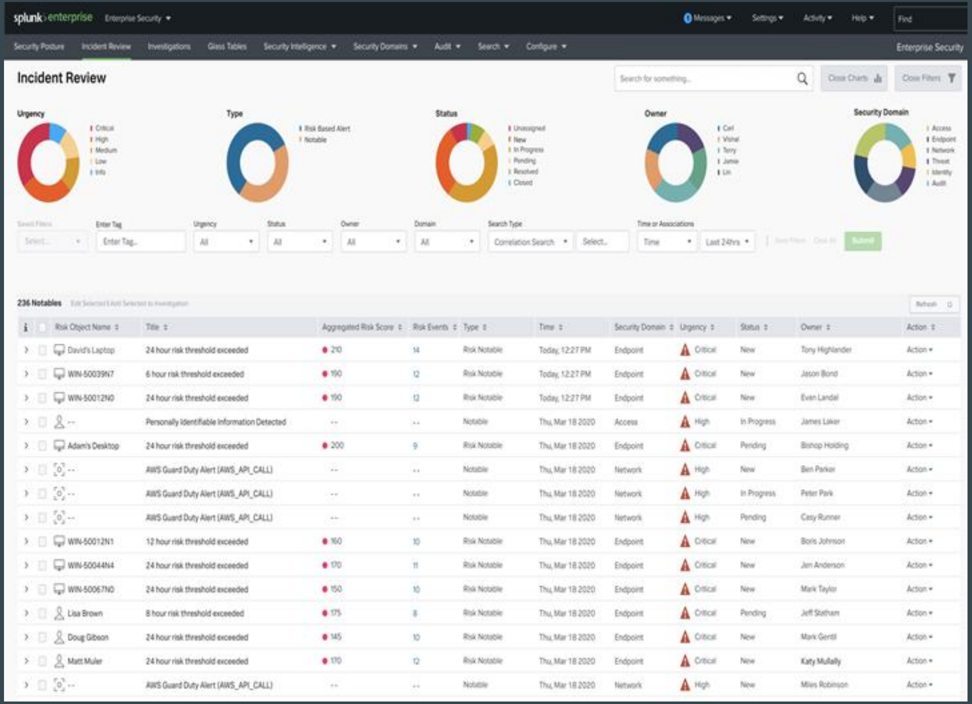
Pros:

- Seamless integration with Microsoft 365
- Minimal on-premise infrastructure required
- Fast deployment and setup

Cons:

- Ongoing cost tied to data volume
- Requires Azure familiarity for customization
- Less control over backend infrastructure

Splunk Enterprise Security



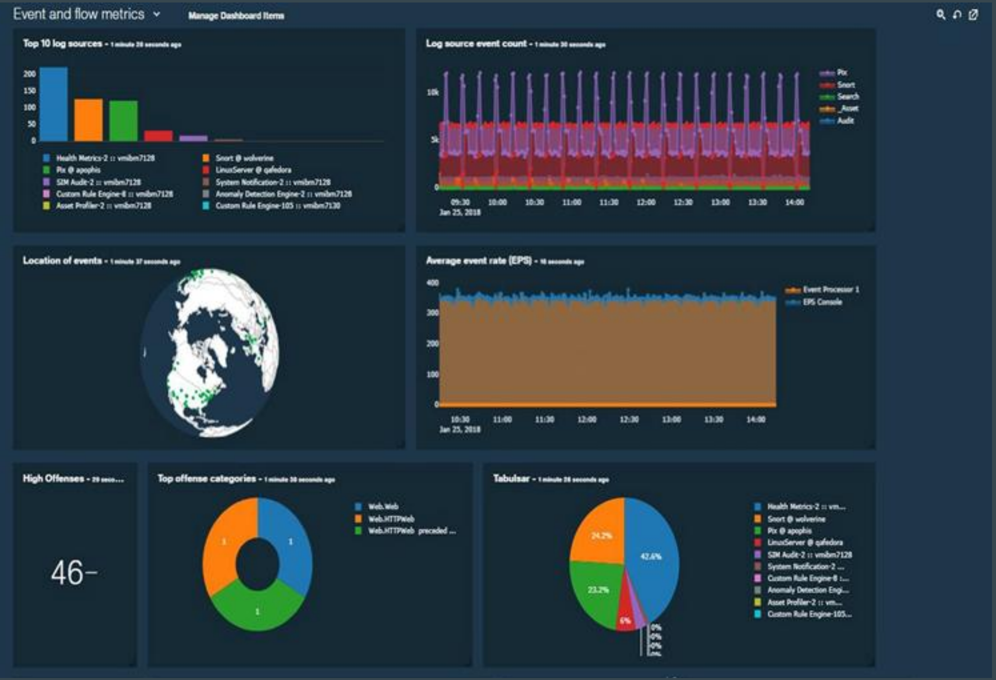
Pros:

- Extremely flexible and scalable
- Strong ecosystem of plugins and integrations
- Ideal for forensic investigation and custom use cases

Cons:

- Higher licensing and infrastructure requirements
- Steeper learning curve for configuration
- Needs dedicated server resources if self-hosted

IBM QRadar



Pros:

- Enterprise-level capabilities
- Excellent compliance support (HIPAA, PCI, etc.)
- Strong Fortinet and Windows log ingestion support

Cons:

- Complex setup and tuning
- Higher upfront cost and resource demands
- Better suited for larger networks

Final SIEM Recommendation

- Primary Recommendation:
 Microsoft Sentinel: Integrates well with Microsoft 365, offers centralized logging, and provides cost-effective cloud-based SIEM for multi-site visibility.

- Alternate Recommendation:
 Splunk Enterprise Security: Ideal if you want full control over your analytics stack and have the staff to manage log ingestion and tuning.

- IBM QRadar is recommended only if your environment requires in-depth compliance auditing and dedicated threat analysis. IBM QRadar is typically more suited for enterprise or government use.

Intrusion Detection & Prevention Systems (IDS/IPS)

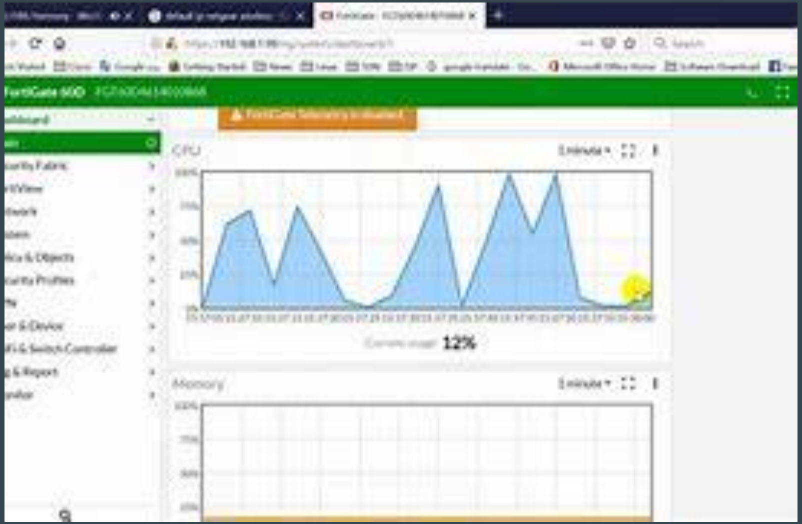
Detects and blocks suspicious activities that could compromise network security.

- Monitor inbound and outbound traffic in real time
- Identify known attack signatures, anomalous behaviors, and policy violations.
- Block threats

3 Devices

- FortiGate Integrated IPS
- Snort
- Suricata

Fortigate Integrated IPS



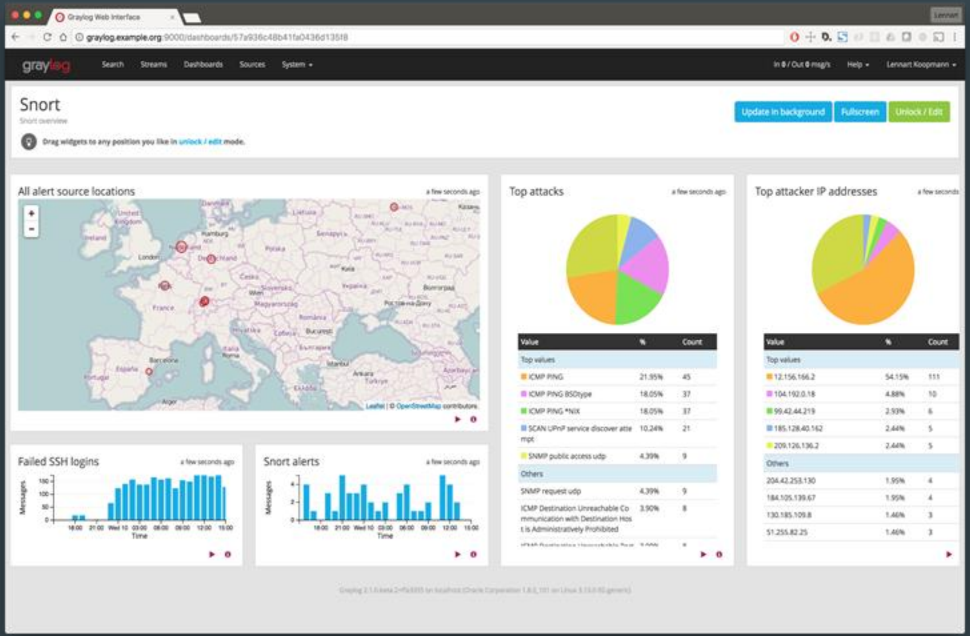
Pros:

- Integrated with Fortigate!
- Simplifies configuration via centralized FortiGate GUI
- Low latency due to hardware acceleration
- Cost-effective inclusion with UTM bundle

Cons:

- Requires annual FortiGuard IPS license for updates
- Limited customization compared to standalone IDS/IPS

Snort



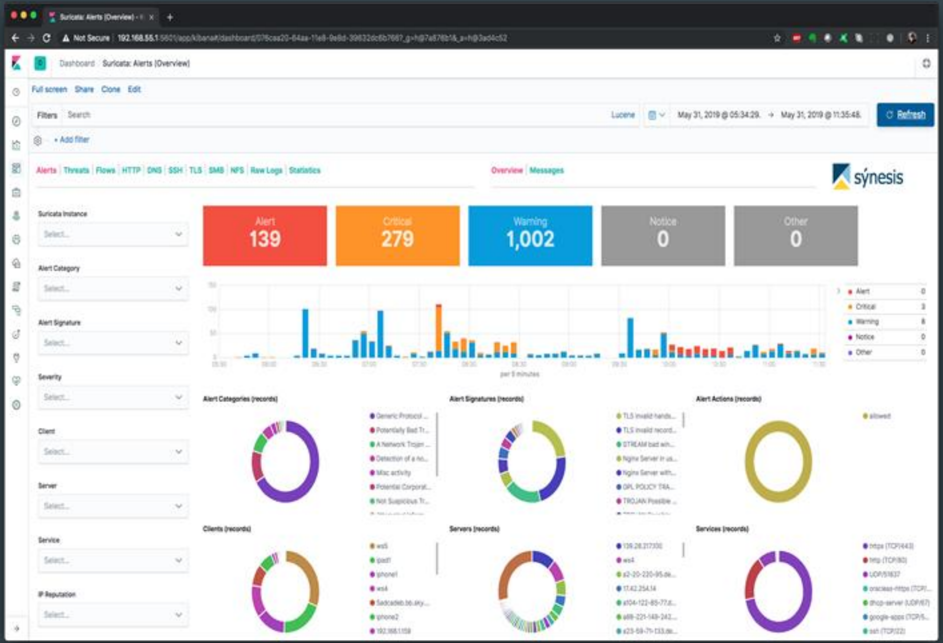
Pros:

- Free and highly customizable
- Large support community
- Lightweight for passive detection

Cons:

- Manual configuration and maintenance required
- No built-in prevention unless used inline
- Requires separate hardware and monitoring interface

Suricata



Pros:

- Excellent performance for open-source solution
- Strong integration with security logging tools
- More modern architecture than Snort

Cons:

- More resource-intensive than Snort
- Requires tuning for accuracy and efficiency
- No GUI by default

Final Intrusion Detection & Prevention Systems (IDS/IPS) Recommendation

- Primary Recommendation:
  Use FortiGate's integrated IPS as part of the existing firewall deployment. It provides centralized control, cost efficiency, and solid protection against known threats with minimal overhead.

- Alternate Recommendation (Advanced or Research Purposes):
  Consider deploying Suricata on a dedicated server if you plan to extend threat monitoring, research traffic behavior, or integrate with a custom SIEM or ELK stack.

- Snort may be useful for educational/demo use but is less practical in this environment without dedicated security staff.

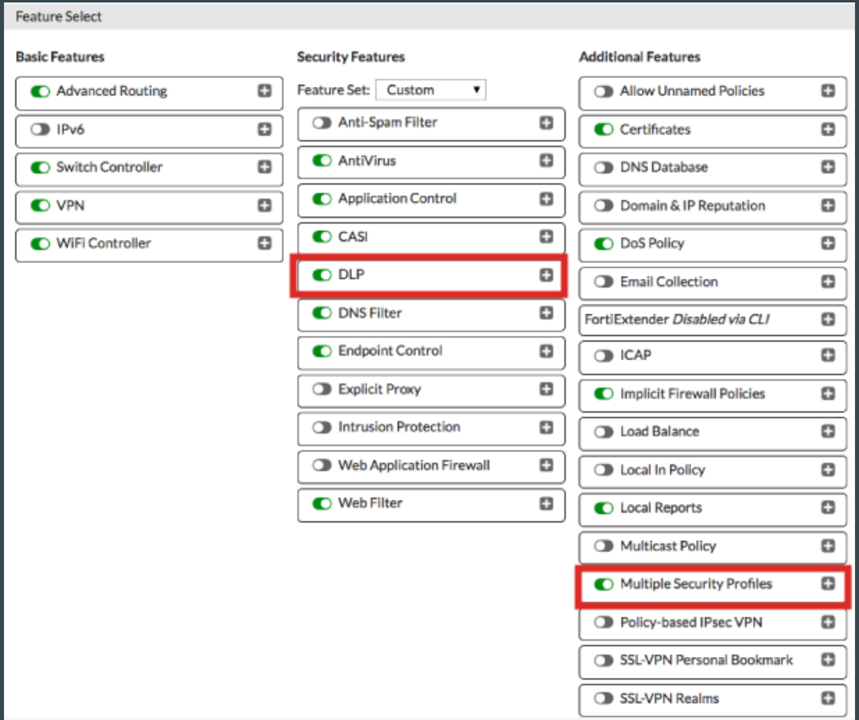
Data Loss Prevention (DLP)

Monitors, detects, and prevents unauthorized transmission or exposure of sensitive data such as PII, financial information, or proprietary documents.

3 Options

- FortiGate Integrated DLP
- Microsoft Purview DLP (Cloud + Endpoint)
- Symantec DLP (Broadcom)

FortiGate Integrated DLP



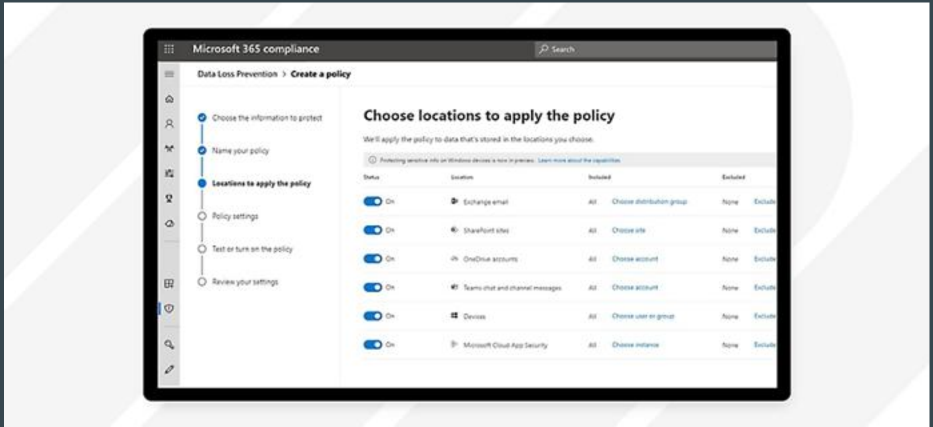
Pros:

- No extra hardware or software needed
- Centralized management via FortiGate GUI
- Integrated into firewall policies and SSL inspection
- Good for small to midsize deployments

Cons:

- Limited to network-layer traffic
- Fewer advanced data classification options
- Lacks endpoint coverage

Microsoft Purview DLP (Cloud + Endpoint)



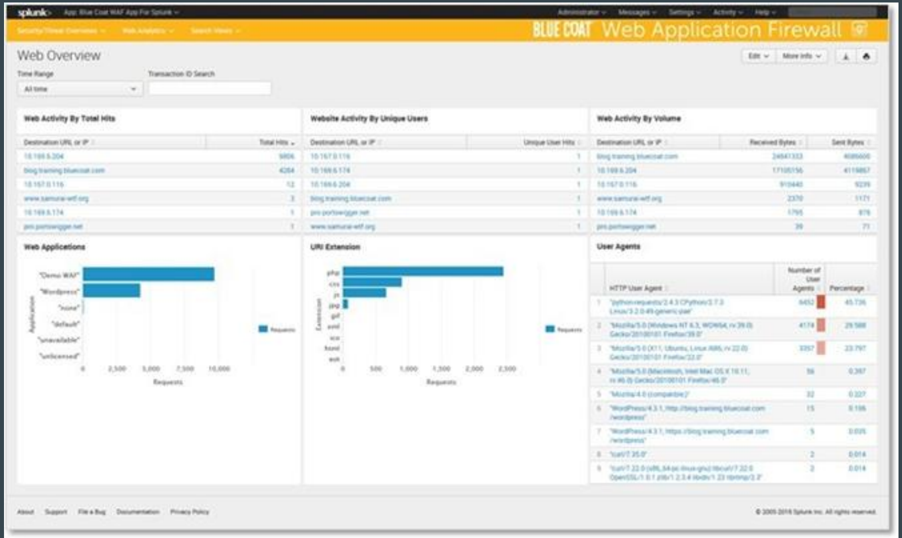
Pros:

- Excellent coverage across cloud and endpoints
- Scalable, policy-driven, and centrally managed
- Provides user education with non-blocking alerts

Cons:

- Requires Microsoft 365 E5 or add-on license
- Cloud-focused — less effective for non-Microsoft systems
- Dependency on Azure configuration

Symantec DLP (Broadcom)



Pros:

- Extremely granular policy and user control
- Enterprise-level scalability and reporting
- Covers all data channels including cloud, network, storage, and endpoints

Cons:

- Complex deployment and maintenance
- Higher cost (licensing + infrastructure)
- Requires trained security/admin staff

Final DLP Recommendation

Primary Recommendation:
Use FortiGate Integrated DLP at all locations for low-cost enforcement of basic data protection policies on network traffic.

Alternate Recommendation:
Consider Microsoft Purview DLP if the organization is fully integrated with Microsoft 365. It enables broader protection across devices, apps, and collaboration tools with minimal setup overhead.

Advanced Enterprise Option:
Symantec DLP is only recommended if the centers evolve into larger institutions handling sensitive educational records, healthcare data, or face regulatory audits requiring full-spectrum DLP.

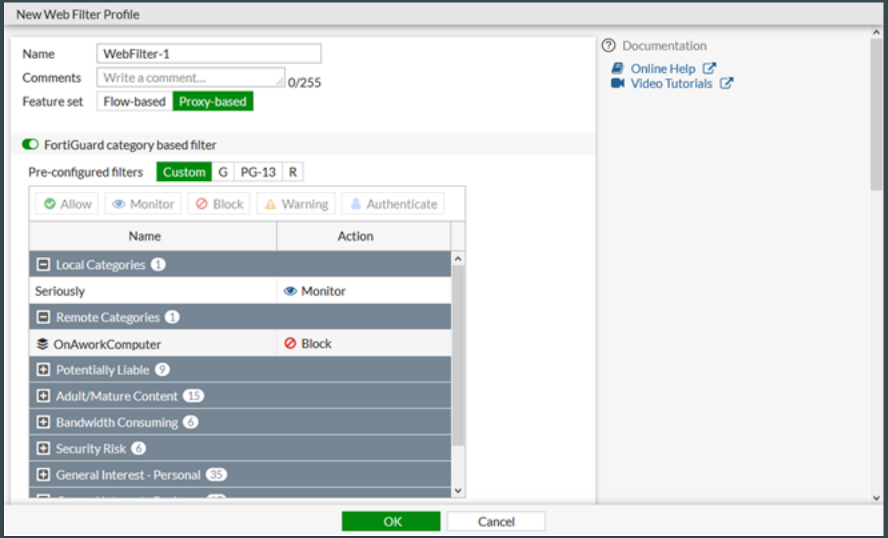
Web Filtering

Controls and monitors users' access to websites and web-based applications to prevent exposure to malicious, inappropriate, or non-work-related content. Web filtering enforces acceptable use policies, reduces bandwidth abuse, and helps prevent phishing and malware infections by blocking risky URLs and domains.

3 Options

- FortiGate Integrated Web Filtering
- Cisco Umbrella
- OpenDNS

FortiGate Integrated Web Filtering



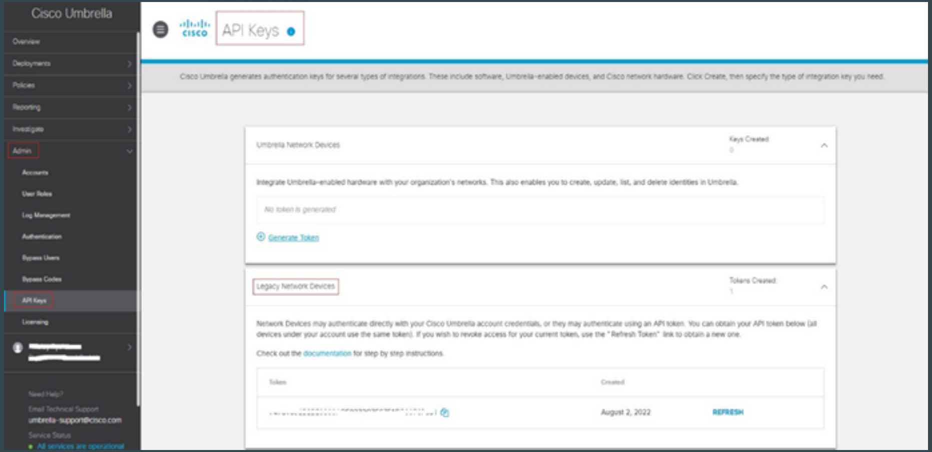
Pros:

- Integrated into FortiGate GUI and policies
- Minimal configuration with centralized management
- Granular control by user, group, time, or application
- Automatically blocks known malicious sites

Cons:

- Requires valid FortiGuard license for category updates
- Limited reporting unless integrated with FortiAnalyzer

Cisco Umbrella



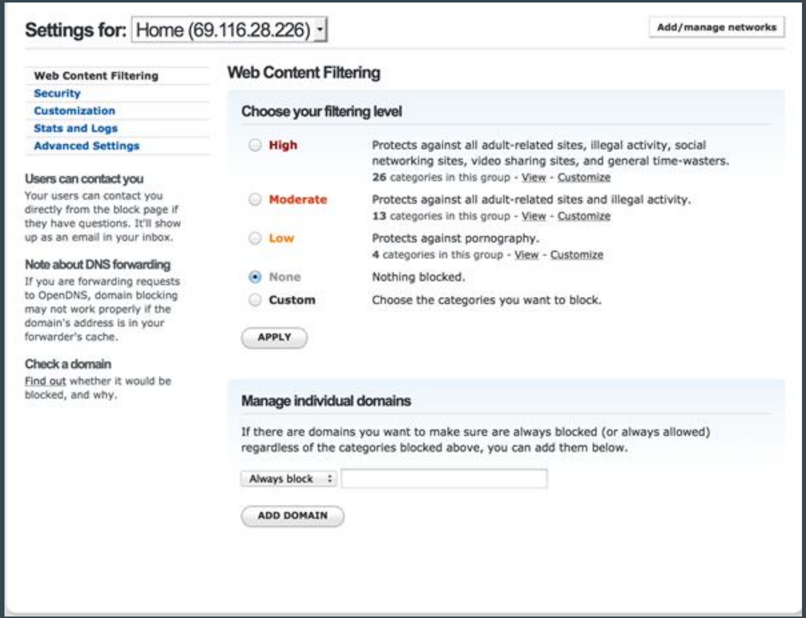
Pros:

- Fast cloud setup
- Protects on-network and off-network users
- Simple per-user policy creation and compliance tracking
- Integrates with Active Directory and SIEM tools

Cons:

- Subscription-based licensing per user/device
- Not ideal for deep packet inspection or fine-grained URL control

OpenDNS



Pros:

- Free for small deployments
- No hardware or agents needed
- Fast performance with global DNS infrastructure
- Easy setup on home or small office networks

Cons:

- Lacks deep URL filtering or SSL inspection
- No user identity or group-based control
- Limited reporting and enforcement capabilities

Final Web Filtering Authentication

Primary Recommendation:
Deploy FortiGate Integrated Web Filtering across all centers. It offers category-based filtering, threat protection, and centralized management with minimal setup.

Alternate Recommendation):
Use Cisco Umbrella if users frequently operate outside the physical network (e.g., hybrid tutoring). It offers flexible, cloud-based protection at the DNS layer with strong policy enforcement.

Entry-Level or Satellite Option:
OpenDNS is a reliable no-cost choice for smaller branch offices, non-critical labs, or to extend basic protection without licensing costs.

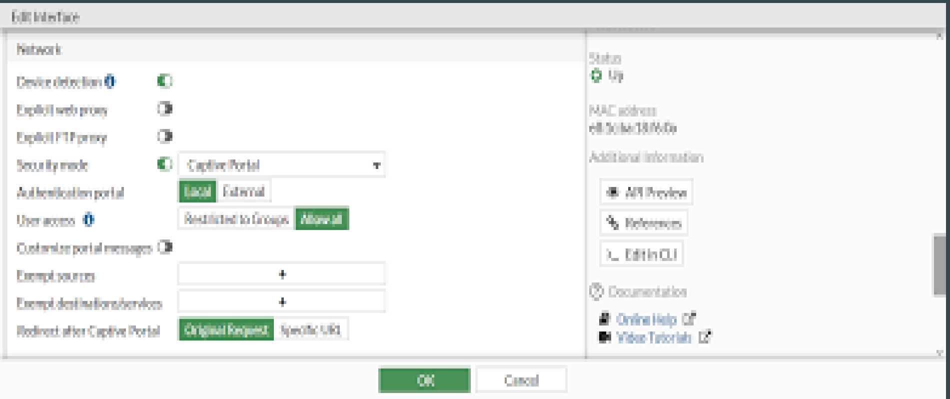
Web Authentication

Controls user access to internal systems and web applications by verifying identity through secure login methods. Web authentication ensures that only authorized users can access protected content and services, while enabling auditing, session control, and integration with directory services like Active Directory or Azure AD.

3 Options

- Fortigate Captive Portal
- Okta
- Apache HTTP Server with LDAP or RADIUS Auth

FortiGate Captive Portal



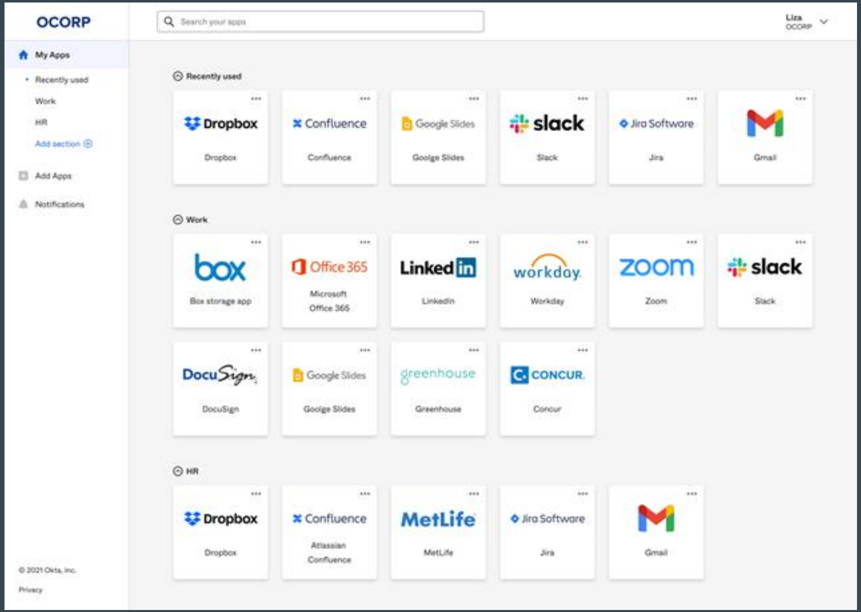
Pros:

- No additional hardware or licensing required
- Simple setup for guest and visitor networks
- Can enforce per-user bandwidth limits and logging
- Integrated with FortiGate policy enforcement

Cons:

- Basic UI and limited customization
- Does not support modern SSO (OAuth2, SAML, etc.)
- Best suited for small- to mid-scale environments

Okta



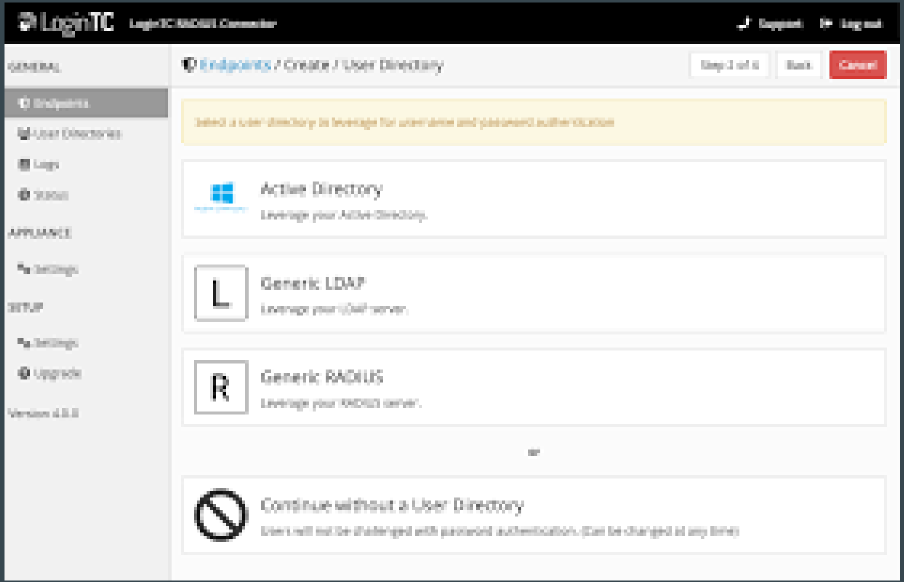
Pros:

- Enhances security and user experience with SSO
- Reduces password fatigue and help desk resets
- Scales well across multiple locations and devices
- Integrates with cloud apps, VPNs, and legacy systems

Cons:

- Subscription-based pricing per user/month
- Requires some configuration for app integrations
- Advanced features may be overkill for small centers

Apache HTTP Server with LDAP or RADIUS Auth



Pros:

- Free and open-source
- Highly customizable for internal portals
- Lightweight solution for legacy or custom apps

Cons:

- Requires server administration and manual configuration
- No native MFA or modern token-based authentication
- Limited scalability and reporting

Final Web Authentication Recommendation

Primary Recommendation:
Use FortiGate Captive Portal for guest and non-SSO user web access. It's cost-effective, simple to deploy, and integrates cleanly into existing firewall policies across all centers.

Alternate Recommendation:
Consider Okta for centralized web authentication and SSO if the tutoring centers expand their use of SaaS platforms, remote staff, or require MFA and secure cloud access.

Custom or Local Option:
Use Apache with LDAP/RADIUS for internal applications that don't integrate with external SSO platforms — suitable for legacy systems and low-traffic administrative portals.

Internet Service Provider (ISP) Data and Bandwidth Calculations

Calculated by estimating the number of users and devices, their usage patterns, and then adding a margin for peak traffic and future growth.

Factors include:

- Device Count & Usage
- Application Requirements
- Peak Traffic Estimation
- Symmetrical Speeds
- Industry Guidelines

Internet Service Provider (ISP) Tutoring Center Estimations

- Estimated 50 devices in use during peak periods
- 5 Mbps of bandwidth for each device during those peak periods
- Baseline need of 250 Mbps
- Recommended adding overhead of 20–30% and planning for growth
- A plan offering 300 Mbps or 500 Mbps symmetrical speeds would provide the best service to the tutoring centers..

Internet Service Provider (ISP) Recommendations

Verizon Business: Best overall reliability, but higher cost

AT&T Business: Best for customer satisfaction

Comcast Business: Best for availability

Recommended ISP Plan

- Romeoville & Chicago: Fiber-optic AT&T Business for high-speed, low-latency connectivity. Romeoville would likely benefit from the higher bandwidth rate.
  - Internet 300: $55 per month plus tax, offering 300 Mbps symmetrical upload/download speeds.
  - Internet 500: $65 per month plus tax, providing 500 Mbps symmetrical speeds.
- Aurora:
  - Without Fiber upgrade: AT&T Business Internet 100. $55 per month plus tax
    - Copper-based connection type, 100 Mbps download, 20 Mbps upload- Asymmetrical speeds
    - Will likely cause connection issues with the estimated 20 + concurrent users.
  - With Fiber upgrade: AT&T Business Fiber
    - Enhance Aurora’s tutoring centers' connectivity.
    - Standard upgrade- installation cost low.
    - If minor construction is needed, price will likely go up.

Equipment Selection and Cost Breakdown

Major recommended cost areas include:

- Core Networking Equipment: Fortinet switches, routers, and firewalls to support segmented VLANs, secure remote access, and future scalability.

  Workstations & End-User Devices: 45 HP laptops for students and staff use, along with headsets, VoIP phones, and printers.
- Software Licensing: 45 Windows 11 Pro licenses to enable security, management, and remote functionality.
- ISP Upgrade: Business Fiber internet at each location for symmetrical bandwidth and high reliability.
- The finalized cost comes to $59,309.54

Equipment Selection and Cost Breakdown

An optional expansion budget of $20,000 has been proposed to further enhance the network and computing environment. If approved, this additional funding would be prioritized for:

- Expanding the laptop pool to support additional staff or remote tutors.
- Adding more VoIP phones or smart displays in instructional areas.
- Upgrading Aurora's internet connection to AT&T Business Fiber for uniform performance across all sites.

Network Documentation Overview

Purpose: Ensures operational continuity, compliance, and technical clarity

- Topology Diagrams:
  - Logical & physical layout of routers, switches, firewalls, APs, VLANs
- Configuration Manuals:
  - Setup & management instructions for FortiGate, FortiSwitch, FortiAP
- Security Policies:
  - SOPPA, FERPA, Illinois PIPA compliance
- Backup & Recovery Plans:
  - Off-site storage, restoration procedures, DR drills

User & Admin Documentation

User Documentation Includes:

- Access guides for VPN, wireless, shared drives
- Troubleshooting steps with escalation pathways
- Printer/file resource access instructions

Admin Documentation Includes:

- Detailed topology with VLAN/IP assignments
- Device configs, firewall rules, SIEM integration
- Scheduled maintenance and security audit protocols

IT Policy Highlights

Acceptable Use Policy (AUP):

- No unauthorized software or illegal activity
- Mandatory compliance with data protection standards

Access Control Policy (ACP):

- Role-Based Access Control (RBAC)
- Multi-factor authentication for sensitive systems

Incident & Disaster Plans (IRP / DRP):

- Rapid containment, post-incident analysis
- Regular backups & defined restoration roles

Change Management Policy (CMP):

- Approval workflows, rollback plans, revision logs

Device Configuration Summary

Router Setup (FortiGate):

- IP assignment, DHCP, firewall & VPN configuration
- Logging integrated with SIEM

Switch Setup (FortiSwitch):

- Static IP, VLAN assignments, STP enabled

Wireless APs (FortiAP):

- WPA3 enterprise SSIDs mapped to VLANs
- Captive portal for guests, QoS for VoIP/video

Ongoing Security & Monitoring

- Microsoft Endpoint Manager: Endpoint control
- SIEM Integration: Real-time monitoring via Graylog/Splunk/ELK
- Patch Management: Scheduled vulnerability scans & updates
- Security Audits: Periodic reviews of firewall rules and access logs

Infrastructure Testing Overview

Connectivity Tests:

- Ping, Traceroute, VLAN testing, VPN validation

Performance Tools:

- iPerf, NetIO, Wireshark, Apache JMeter

Redundancy Checks:

- STP active, switch failover simulation
- VPN tunnels tested between all sites

Wireless Heatmap Planning

- Coverage tested using Ekahau / NetSpot
- SSIDs segregated by VLAN: Staff, Tutors, Guest
- Romeoville: 3 APs centrally placed for full coverage
- Chicago & Aurora: APs placed near high-traffic areas
- Signal optimized for drywall & office layout obstacles

Load Balancing & Traffic Monitoring

- Romeoville: Balanced traffic across DMZ, Staff, Guest VLANs
- Chicago: Dual-switch with guest network strain mitigation
- Aurora: Stable with <70% utilization under load

Tools Used:

- Wireshark, JMeter, Nagios, PRTG

Bandwidth Estimates by Location

Site	Est. Devices	Peak Bandwidth	Recommended ISP Plan
Romeoville	~40	260 Mbps	300–500 Mbps
Chicago	~33	215 Mbps	300 Mbps
Aurora	~25	165 Mbps	300 Mbps

- *30% headroom included for scaling*
- *All sites tested for packet loss & latency*

Conclusion

- Robust documentation enables scalability & security
- All policies aligned with legal and institutional standards
- Testing confirms infrastructure resilience